



VIT-AP
UNIVERSITY

Module 1: Understanding Cloud Computing

Risks and Challenges

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OUTLINE

- Risks and Challenges
 - **Introduction to the risks and challenges associated with cloud computing.**
 - **Detailed discussion on security and privacy concerns.**
 - **Examination of reliability and downtime issues.**
 - **Exploration of legal and regulatory challenges.**
 - **Strategies for managing and mitigating risks.**
- Summary
- Quiz
- References

- Data Breaches
 - Definition: Unauthorized access to sensitive data.
 - Examples: High-profile data breaches (e.g., Equifax, Target).
 - Consequences: Financial loss, reputational damage, legal repercussions.
- Compliance Issues
 - Regulations: GDPR General Data Protection Regulation, European Union, HIPAA Health Insurance Portability and Accountability Act, United States, CCPA California Consumer Privacy Act California, USA, etc.
 - Challenges: Ensuring cloud services comply with industry standards and regulations.
 - Penalties: Non-compliance can result in fines and legal actions.
 - Key Points
 - Importance of data encryption (in transit and at rest).
 - Implementing robust access controls and authentication mechanisms.
 - Regular security audits and vulnerability assessments.

Reliability and Downtime

- Reliability and Downtime
 - Service Outages
 - Examples: AWS, Azure, Google Cloud outages.
 - Impact: Business operations halted, financial losses, customer dissatisfaction.
 - Causes: Hardware failures, software bugs, network issues.
 - Dependency on Providers
 - Vendor Lock-in: Difficulty in switching providers due to proprietary services and data formats.
 - Service Level Agreements (SLAs): Understanding and negotiating SLAs to ensure uptime guarantees.
 - Redundancy and Failover Strategies: Importance of having backup plans and failover mechanisms.
 - Key Points
 - Choosing providers with strong track records and reliable infrastructure.
 - Diversifying cloud providers to reduce dependency.
 - Regularly reviewing and updating disaster recovery plans.

- ❑ Security and Privacy
- ❑ Reliability and Downtime
- ❑ Legal and Regulatory Challenges

Legal and Regulatory Challenges

- Legal and Regulatory Challenges

- Data Sovereignty

- Definition: Data is subject to the laws and governance structures within the nation it is collected.
 - Challenges: Cross-border data transfer regulations, varying international laws.
 - Impact: Compliance with local laws can complicate global operations.

- Legal Compliance

- Requirements: Adhering to laws like GDPR, HIPAA, and other regional regulations.
 - Challenges: Keeping up with changing regulations and ensuring ongoing compliance.
 - Penalties: Severe fines, legal actions, and operational restrictions.

- Key Points

- Understanding the legal landscape of data storage and processing locations.
 - Implementing data classification and protection policies.
 - Engaging legal expertise to navigate complex regulatory environments.

- ❑ Security and Privacy
- ❑ Reliability and Downtime
- ❑ Legal and Regulatory Challenges
- ❑ Managing Risks

Managing Risks

- Managing Risks

- Best Practices for Mitigating Risks

- Security Measures: Regular updates, patch management, intrusion detection systems.
 - Data Protection: Encryption, backup and recovery plans, data masking.
 - Access Control: Implementing least privilege access, multi-factor authentication.
 - Monitoring and Auditing: Continuous monitoring of cloud environments, regular security audits.
 - Vendor Management: Thoroughly vetting cloud providers, understanding their security measures and compliance standards.
 - Training and Awareness: Regular training for employees on security best practices and phishing awareness.

Summary on Basic Concepts and Terminology, Goals and Benefits

- ❑ Introduction to the risks and challenges associated with cloud computing.
- ❑ Detailed discussion on security and privacy concerns.
 - ❑ Data Breaches
 - ❑ Compliance Issues
- ❑ Examination of reliability and downtime issues.
 - ❑ Service Outages
 - ❑ Dependency on Providers
- ❑ Exploration of legal and regulatory challenges.
 - ❑ Data Sovereignty
 - ❑ Legal Compliance
- ❑ Strategies for managing and mitigating risk
 - ❑ Security Measures
 - ❑ Data Protection
 - ❑ Access Control
 - ❑ Monitoring and Auditing
 - ❑ Vendor Management
 - ❑ Training and Awareness

Quiz

- **Q. What does SLA stand for in cloud computing?**
 - **A) Service Level Agreement**
 - **B) Storage Level Agreement**
 - **C) Service Latency Agreement**
 - **D) Security Level Agreement**
 - **Answer: A) Service Level Agreement**
- **Q: Which of the following is a major security concern in cloud computing?**
 - **A) Low bandwidth**
 - **B) Data breaches**
 - **C) High latency**
 - **D) Scalability issues**
 - **Answer: B) Data breaches**
- **Q: What is the primary cause of service outages in cloud environments?**
 - **A) Unlimited storage capacity**
 - **B) Hardware failures, software bugs, and network issues**
 - **C) Constant uptime**
 - **D) High availability**
 - **Answer: B) Hardware failures, software bugs, and network issues**
- **Describe the major security concerns associated with cloud computing and provide examples of how these concerns can be mitigated**

References

1. Thomas Erl, Ricardo Puttini, Zaigham Mahmood, “Cloud Computing: Concepts, Technology & Architecture”, PHI Publications, 2013.

Questions



**Thank
You**